



NATIONAL INSTITUTE OF BUSINESS MANAGEMENT

BSc (Hons) Computing|ITB|EHN|Year-1-19.1P

DIPLOMA IN SOFTWARE ENGINEERING-19.1P

Computer Technology NIB-SOC-COM-1-3-05

19 October 2019

TIME –2 HOURS

Instructions : Answer all questions

Question 01 (Marks 32: 16 x 2)

1. Resistors that have set resistance, are known as
 - A. fixed resistors
 - B. variable resistors
 - C. thermistors
 - D. fixators
2. Current through each resistor when they are connected in series is
 - A. different
 - B. same
 - C. can be both A and B
 - D. decreasing
3. Three resistances $20\ \Omega$, $30\ \Omega$ and $60\ \Omega$ are connected in parallel, their combined resistance is given by
 - A. $110\ \Omega$
 - B. $50\ \Omega$
 - C. $20\ \Omega$
 - D. $10\ \Omega$
4. Combined resistance of $5\ \Omega$ and $10\ \Omega$ is equal to
 - A. $10\ \Omega$
 - B. $16\ \Omega$
 - C. $15\ \Omega$
 - D. $20\ \Omega$

5. Two $10\ \Omega$ resistors are connected in parallel, their equivalent resistance is
- A. $5\ \Omega$
 - B. $0.2\ \Omega$
 - C. $15\ \Omega$
 - D. $20\ \Omega$
6. Ammeter should always have a
- A. high resistance
 - B. low resistance
 - C. low voltage
 - D. high voltage
7. If 2 resistors of $5\text{k}\Omega$ and $7\text{k}\Omega$ are connected in series across a 15V battery, then current flowing through each of resistance should be
- A. 2mA
 - B. 3.6mA
 - C. 5mA
 - D. 1.25mA
8. If 2 resistors of $5\text{k}\Omega$ and $7\text{k}\Omega$ are connected in series across a 15V battery, then potential difference across each of resistance should be
- A. 2V , 3V
 - B. 10V , 14V
 - C. 5V , 7V
 - D. 8V , 6V
9. Device that is used to store charge, is named as
- A. capacitor
 - B. resistor
 - C. transistor
 - D. diode
10. If the input frequency of a half wave rectifier is 100Hz , then the ripple frequency will be_____
- a) 150Hz
 - b) 200Hz
 - c) 100Hz
 - d) 300Hz

11. If input frequency is 50Hz for a full wave rectifier, the ripple frequency of it would be _____

- a) 100Hz
- b) 50Hz
- c) 25Hz
- d) 500Hz

12. In parts of the processor, adders are used to calculate _____

- a) Addresses
- b) Table indices
- c) Increment and decrement operators
- d) All of the Mentioned

13. Total number of inputs in a half adder is _____

- a) 2
- b) 3
- c) 4
- d) 1

14. In which operation carry is obtained?

- a) Subtraction
- b) Addition
- c) Multiplication
- d) Both addition and subtraction

15. If A and B are the inputs of a half adder, the sum is given by _____

- a) A AND B
- b) A OR B
- c) A XOR B
- d) A EX-NOR B

16. If A and B are the inputs of a half adder, the carry is given by _____

- a) A AND B
- b) A OR B
- c) A XOR B
- d) A EX-NOR B

Question 02 (25 Marks)

Half adders are used to add 2, 1bit binary numbers.as the two inputs A & B will generate a output which known as carry and sum.

a.1) Fill the binary values output (carry & Sum) for given input A & B (Marks 8)

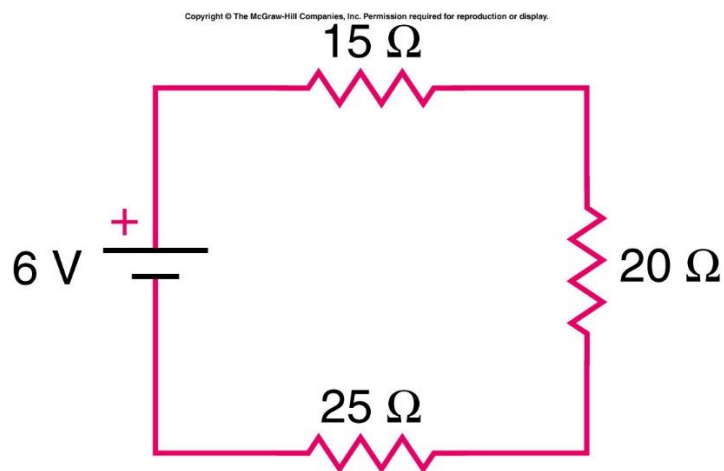
Input		output	
A	B	Carry	Sum
0	0		
0	1		
1	0		
1	1		

a.2) Generate separate K-maps for carry & Sum (Marks 10)

a.3) Obtain the Boolean expressions for SUM & CARRY (Marks 4)

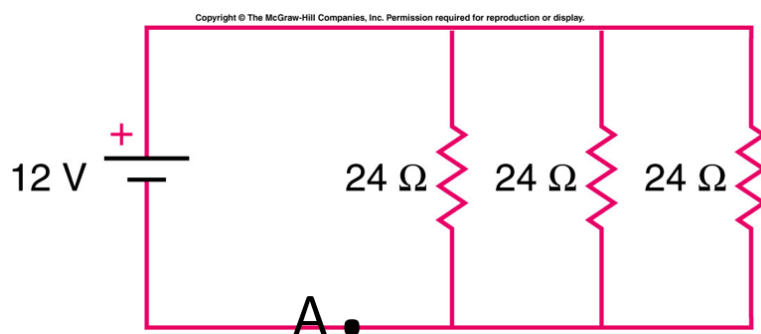
a.4) Draw the Logical circuit diagram of the half adder as per the Obtained Boolean Expression at question number (a.3) (Marks 3)

Question 03 (Marks 16)



b.1) Calculate the Total resistance of the above circuit.

b.2) Calculate the current that flow of the above Circuit.



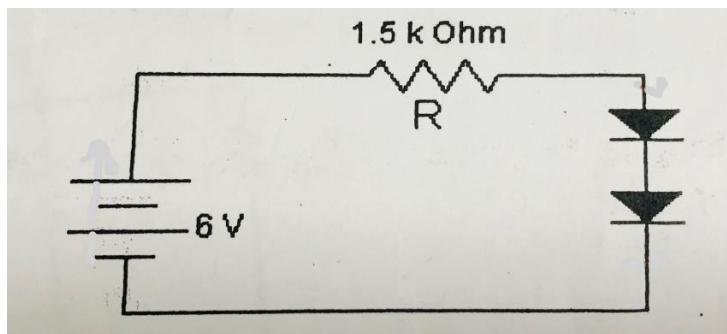
b.1) Calculate the Total resistance of the above circuit.

b.2) What is the current that flows in this circuit at point A?

Question 04 (Marks 5)

c.1) Determine whether the diodes (Si) in the circuit below are forward biased or reverses biased.

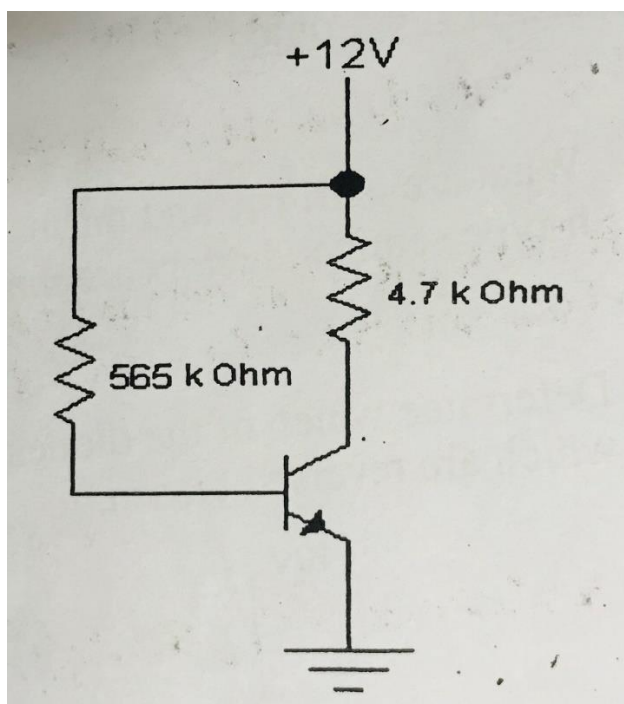
c.2) apply ohms law to the resister R and find out the current flowing through it. (Si=0.7v)



Question 05 (Marks 12)

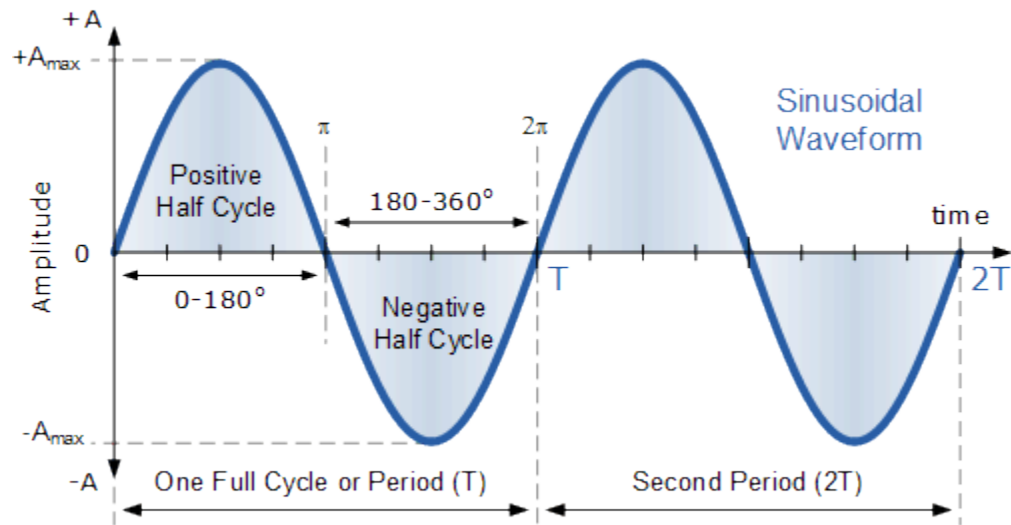
In the circuit shows below find,

- I. V_{CE} when $I_C = 2.5\text{mA}$
- II. I_C when $V_{CE} = 2.6\text{V}$
- III. V_{CE} when $I_C = 0$

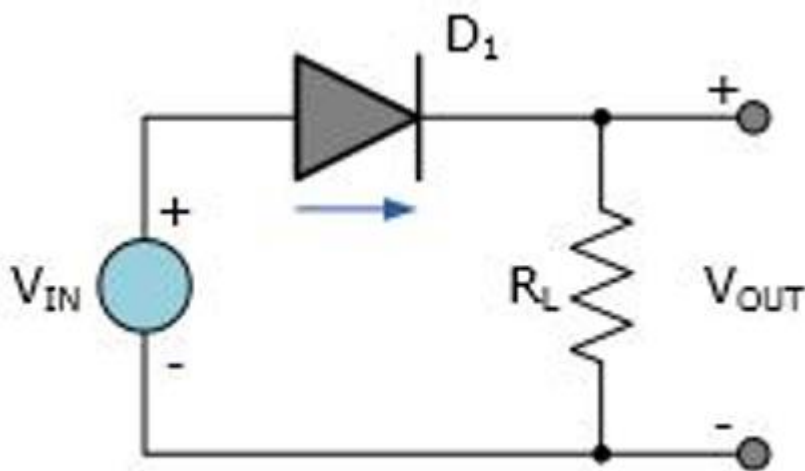


Question 06 (Marks 10)

Use below sinusoidal waveform for question no (e.1) & (e.2)



E.1) Draw the output wave at (V out) according to the given Sinusoidal waveform input at V_{in}



E.2) Draw the output wave at (V out) according to the given Sinusoidal waveform input at V_{ac}

